

5.2. Imbalanced Data

In the last lesson, we prepared the data.

In this lesson, we're going to explore some of the features of the dataset, use visualizations to help us understand those features, and develop a model that solves the problem of imbalanced data by under- and over-sampling.

```
In [1]: import gzip
import json
import pickle

import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from imblearn.over_sampling import RandomOverSampler
from imblearn.under_sampling import RandomUnderSampler
from IPython.display import VimeoVideo
from sklearn.impute import SimpleImputer
from sklearn.metrics import ConfusionMatrixDisplay
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.tree import DecisionTreeClassifier
```

```
In [ ]: VimeoVideo("694058667", h="44426f200b", width=600)
```

Prepare Data

Import

As always, we need to begin by bringing our data into the project, and the function we developed in the previous module is exactly what we need.

```
In [ ]: VimeoVideo("694058628", h="00b4cfd027", width=600)
```

Task 5.2.1: Complete the `wrangle` function below using the code you developed in the last lesson. Then use it to import `poland-bankruptcy-data-2009.json.gz` into the DataFrame `df`.

- [Write a function in Python.](#)

```
In [2]: def wrangle(filename):
# Open compressed file, load into dictionary
with gzip.open(filename, "r") as f:
    data = json.load(f)
# Load dictionary into DataFrame, set index
df = pd.DataFrame().from_dict(data["data"]).set_index("company_id")
return df
```

```
In [3]: df = wrangle("data/poland-bankruptcy-data-2009.json.gz")
print(df.shape)
df.head()
```

(9977, 65)

```
Out[3]:
```

	feat_1	feat_2	feat_3	feat_4	feat_5	feat_6	feat_7	feat_8	feat_9
company_id									
1	0.174190	0.41299	0.14371	1.3480	-28.9820	0.60383	0.219460	1.12250	1.12250
2	0.146240	0.46038	0.28230	1.6294	2.5952	0.00000	0.171850	1.17210	1.17210
3	0.000595	0.22612	0.48839	3.1599	84.8740	0.19114	0.004572	2.98810	2.98810
5	0.188290	0.41504	0.34231	1.9279	-58.2740	0.00000	0.233580	1.40940	1.40940
6	0.182060	0.55615	0.32191	1.6045	16.3140	0.00000	0.182060	0.79808	0.79808

5 rows × 65 columns



Explore

Let's take a moment to refresh our memory on what's in this dataset. In the last lesson, we noticed that the data was stored in a JSON file (similar to a Python dictionary), and we explored the key-value pairs. This time, we're going to look at what the values in those pairs actually are.

```
In [ ]: VimeoVideo("694058591", h="8fc20629aa", width=600)
```

Task 5.2.2: Use the `info` method to explore `df`. What type of features does this dataset have? Which column is the target? Are there columns with missing values that we'll need to address?

- Inspect a DataFrame using the `shape`, `info`, and `head` in pandas.

```
In [4]: # Inspect DataFrame
df.info()
```

<class 'pandas.core.frame.DataFrame'>

Int64Index: 9977 entries, 1 to 10503

Data columns (total 65 columns):

#	Column	Non-Null Count	Dtype
0	feat_1	9977 non-null	float64
1	feat_2	9977 non-null	float64
2	feat_3	9977 non-null	float64
3	feat_4	9960 non-null	float64
4	feat_5	9952 non-null	float64
5	feat_6	9977 non-null	float64
6	feat_7	9977 non-null	float64
7	feat_8	9964 non-null	float64
8	feat_9	9974 non-null	float64
9	feat_10	9977 non-null	float64
10	feat_11	9977 non-null	float64
11	feat_12	9960 non-null	float64
12	feat_13	9935 non-null	float64
13	feat_14	9977 non-null	float64
14	feat_15	9970 non-null	float64
15	feat_16	9964 non-null	float64
16	feat_17	9964 non-null	float64
17	feat_18	9977 non-null	float64
18	feat_19	9935 non-null	float64
19	feat_20	9935 non-null	float64
20	feat_21	9205 non-null	float64
21	feat_22	9977 non-null	float64
22	feat_23	9935 non-null	float64
23	feat_24	9764 non-null	float64
24	feat_25	9977 non-null	float64
25	feat_26	9964 non-null	float64
26	feat_27	9312 non-null	float64
27	feat_28	9765 non-null	float64
28	feat_29	9977 non-null	float64
29	feat_30	9935 non-null	float64
30	feat_31	9935 non-null	float64
31	feat_32	9881 non-null	float64
32	feat_33	9960 non-null	float64
33	feat_34	9964 non-null	float64
34	feat_35	9977 non-null	float64
35	feat_36	9977 non-null	float64
36	feat_37	5499 non-null	float64
37	feat_38	9977 non-null	float64
38	feat_39	9935 non-null	float64
39	feat_40	9960 non-null	float64
40	feat_41	9787 non-null	float64
41	feat_42	9935 non-null	float64
42	feat_43	9935 non-null	float64
43	feat_44	9935 non-null	float64
44	feat_45	9416 non-null	float64
45	feat_46	9960 non-null	float64
46	feat_47	9896 non-null	float64
47	feat_48	9977 non-null	float64
48	feat_49	9935 non-null	float64
49	feat_50	9964 non-null	float64
50	feat_51	9977 non-null	float64
51	feat_52	9896 non-null	float64
52	feat_53	9765 non-null	float64
53	feat_54	9765 non-null	float64
54	feat_55	9977 non-null	float64

```
55 feat_56 9935 non-null float64
56 feat_57 9977 non-null float64
57 feat_58 9948 non-null float64
58 feat_59 9977 non-null float64
59 feat_60 9415 non-null float64
60 feat_61 9961 non-null float64
61 feat_62 9935 non-null float64
62 feat_63 9960 non-null float64
63 feat_64 9765 non-null float64
64 bankrupt 9977 non-null bool
dtypes: bool(1), float64(64)
memory usage: 5.0 MB
```

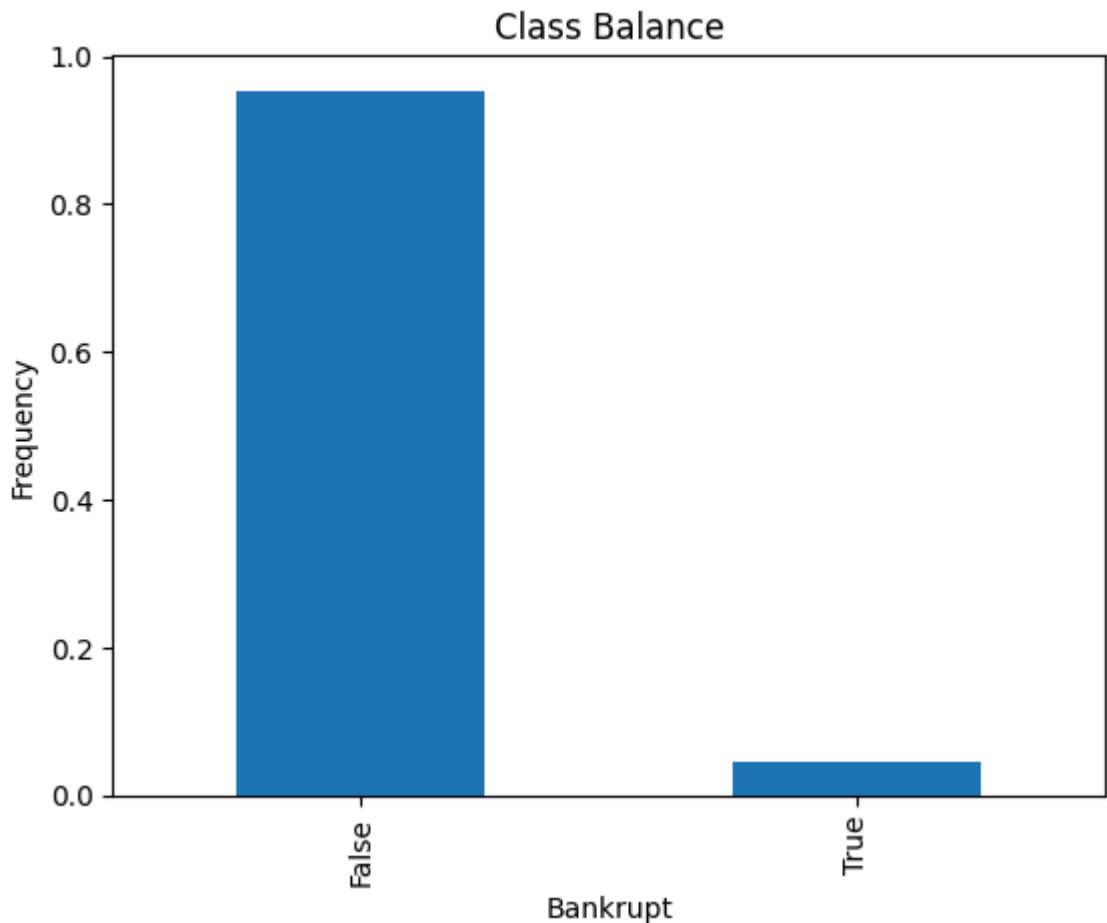
That's solid information. We know all our features are numerical and that we have missing data. But, as always, it's a good idea to do some visualizations to see if there are any interesting trends or ideas we should keep in mind while we work. First, let's take a look at how many firms are bankrupt, and how many are not.

```
In [ ]: VimeoVideo("694058537", h="01caf9ae83", width=600)
```

Task 5.2.3: Create a bar chart of the value counts for the "bankrupt" column. You want to calculate the relative frequencies of the classes, not the raw count, so be sure to set the `normalize` argument to `True`.

- [What's a bar chart?](#)
- [What's a majority class?](#)
- [What's a minority class?](#)
- [What's a positive class?](#)
- [What's a negative class?](#)
- [Aggregate data in a Series using `value\_counts` in pandas.](#)
- [Create a bar chart using pandas.](#)

```
In [5]: # Plot class balance
df["bankrupt"].value_counts(normalize=True).plot(
    kind="bar",
    xlabel="Bankrupt",
    ylabel="Frequency",
    title="Class Balance"
);
```



That's good news for Poland's economy! Since it looks like most of the companies in our dataset are doing all right for themselves, let's drill down a little farther. However, it also shows us that we have an imbalanced dataset, where our majority class is far bigger than our minority class.

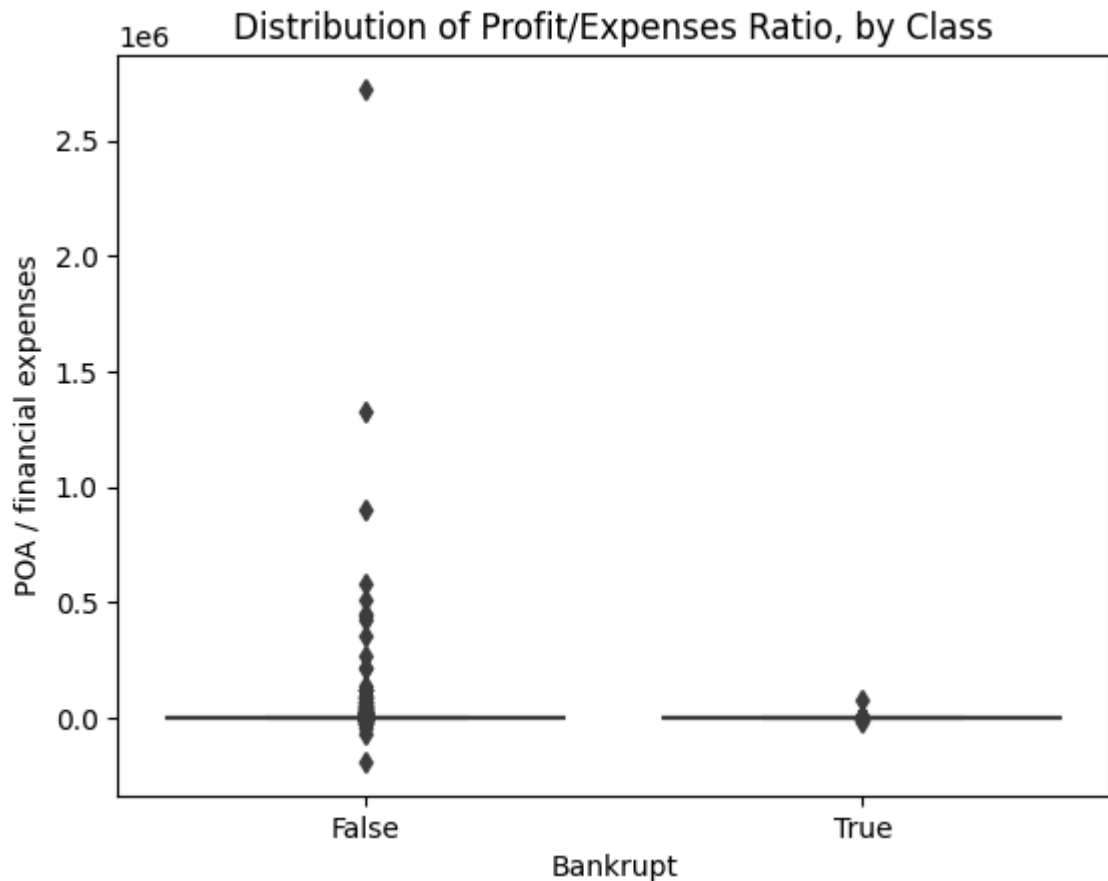
In the last lesson, we saw that there were 64 features of each company, each of which had some kind of numerical value. It might be useful to understand where the values for one of these features cluster, so let's make a boxplot to see how the values in "feat_27" are distributed.

```
In [ ]: VimeoVideo("694058487", h="6e066151d9", width=600)
```

Task 5.2.4: Use seaborn to create a boxplot that shows the distributions of the "feat_27" column for both groups in the "bankrupt" column. Remember to label your axes.

- [What's a boxplot?](#)
- [Create a boxplot using Matplotlib.](#)

```
In [6]: # Create boxplot
sns.boxplot(x="bankrupt", y="feat_27", data=df)
plt.xlabel("Bankrupt")
plt.ylabel("POA / financial expenses")
plt.title("Distribution of Profit/Expenses Ratio, by Class");
```



Why does this look so funny? Remember that boxplots exist to help us see the quartiles in a dataset, and this one doesn't really do that. Let's check the distribution of "feat_27" to see if we can figure out what's going on here.

```
In [ ]: VimeoVideo("694058435", h="8f0ae805d6", width=600)
```

Task 5.2.5: Use the `describe` method on the column for "feat_27". What can you tell about the distribution of the data based on the mean and median?

```
In [7]: # Summary statistics for `feat_27`
df["feat_27"].describe()
```

```
Out[7]: count    9.312000e+03
mean      1.206467e+03
std       3.547726e+04
min       -1.901300e+05
25%       0.000000e+00
50%       1.064300e+00
75%       4.788275e+00
max       2.723000e+06
Name: feat_27, dtype: float64
```

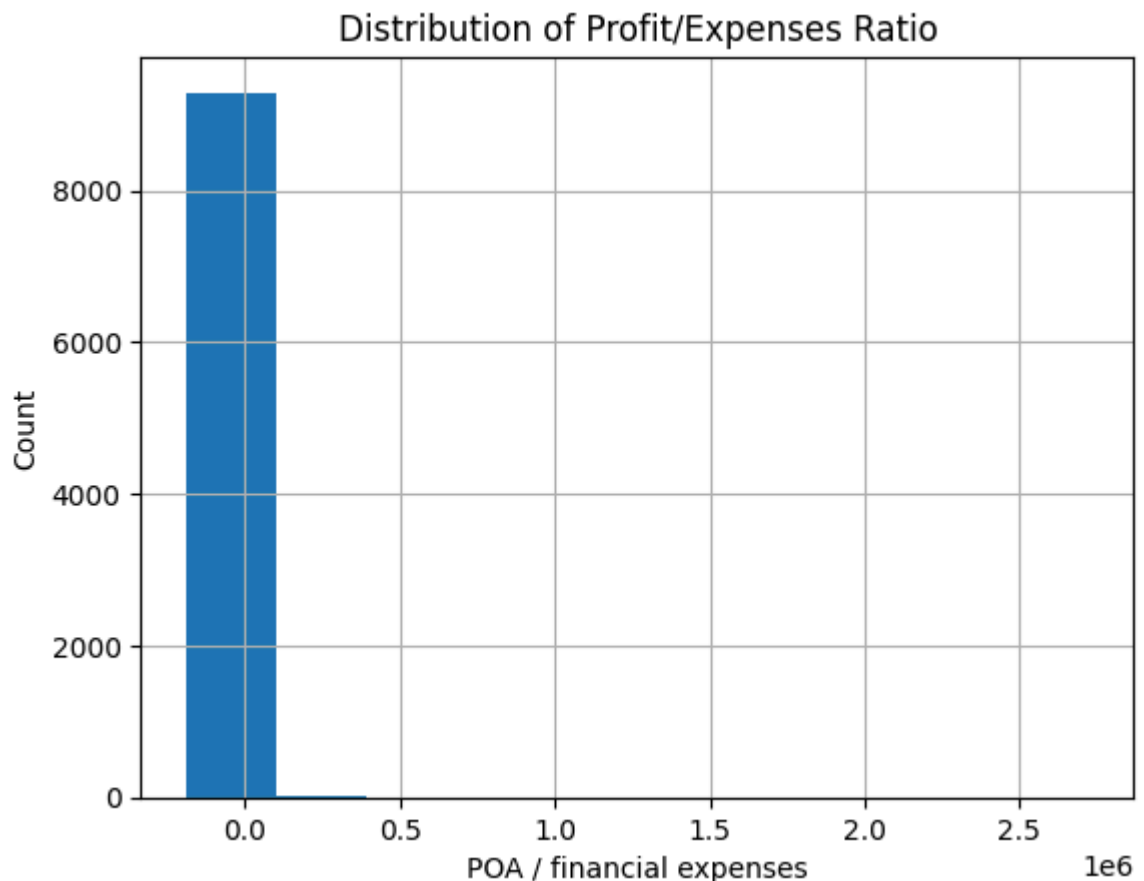
Hmmm. Note that the median is around 1, but the mean is over 1000. That suggests that this feature is skewed to the right. Let's make a histogram to see what the distribution actually looks like.

```
In [ ]: VimeoVideo("694058398", h="1078bb6d8b", width=600)
```

Task 5.2.6: Create a histogram of "feat_27". Make sure to label x-axis "POA / financial expenses", the y-axis "Count", and use the title "Distribution of Profit/Expenses Ratio".

- [What's a histogram?](#)
- [Create a histogram using Matplotlib.](#)

```
In [8]: # Plot histogram of `feat_27`  
df["feat_27"].hist()  
plt.xlabel("POA / financial expenses")  
plt.ylabel("Count"),  
plt.title("Distribution of Profit/Expenses Ratio");
```



Aha! We saw it in the numbers and now we see it in the histogram. The data is very skewed. So, in order to create a helpful boxplot, we need to trim the data.

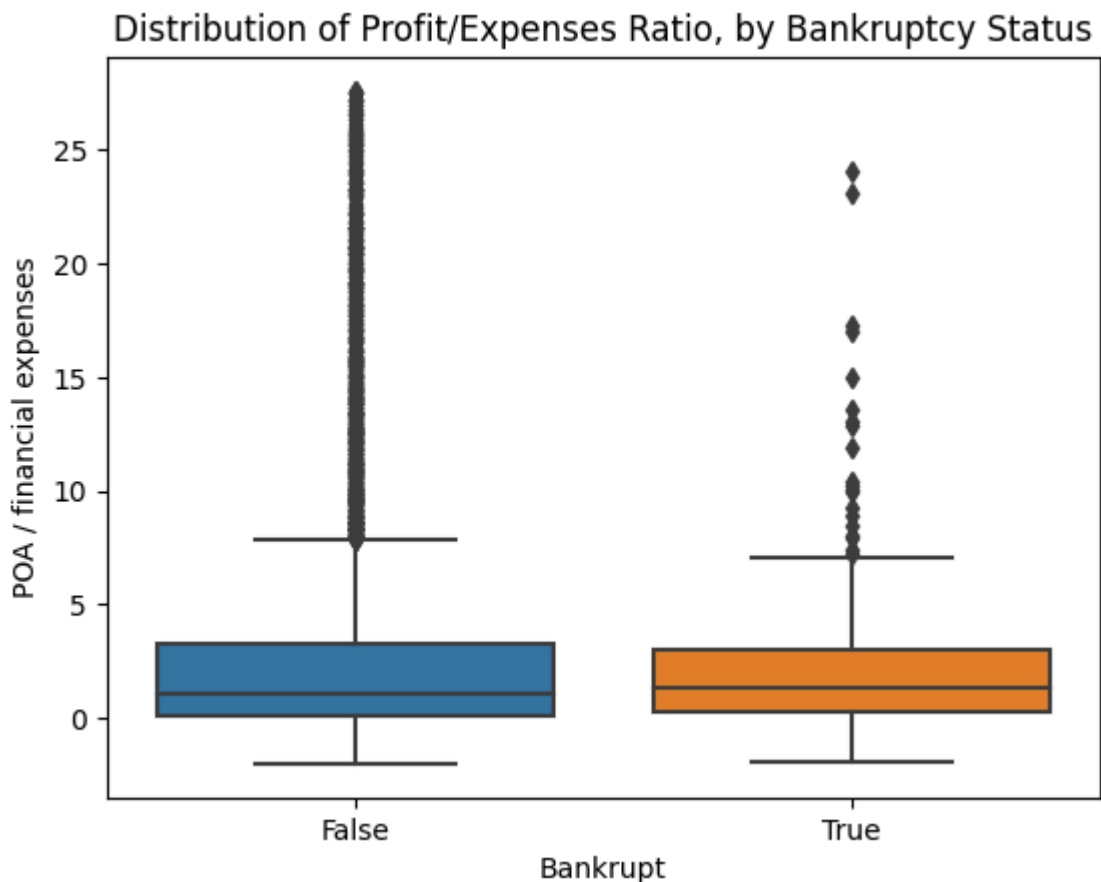
```
In [ ]: VimeoVideo("694058328", h="4aecdc442d", width=600)
```

Task 5.2.7: Recreate the boxplot that you made above, this time only using the values for "feat_27" that fall between the 0.1 and 0.9 quantiles for the column.

- [What's a boxplot?](#)
- [What's a quantile?](#)
- [Calculate the quantiles for a Series in pandas.](#)
- [Create a boxplot using Matplotlib.](#)

```
In [9]: # Create clipped boxplot  
q1, q9 = df["feat_27"].quantile([0.1, 0.9])
```

```
mask = df["feat_27"].between(q1, q9)
sns.boxplot(x="bankrupt", y="feat_27", data=df[mask])
plt.xlabel("Bankrupt")
plt.ylabel("POA / financial expenses")
plt.title("Distribution of Profit/Expenses Ratio, by Bankruptcy Status");
```



That makes a lot more sense. Let's take a look at some of the other features in the dataset to see what else is out there.

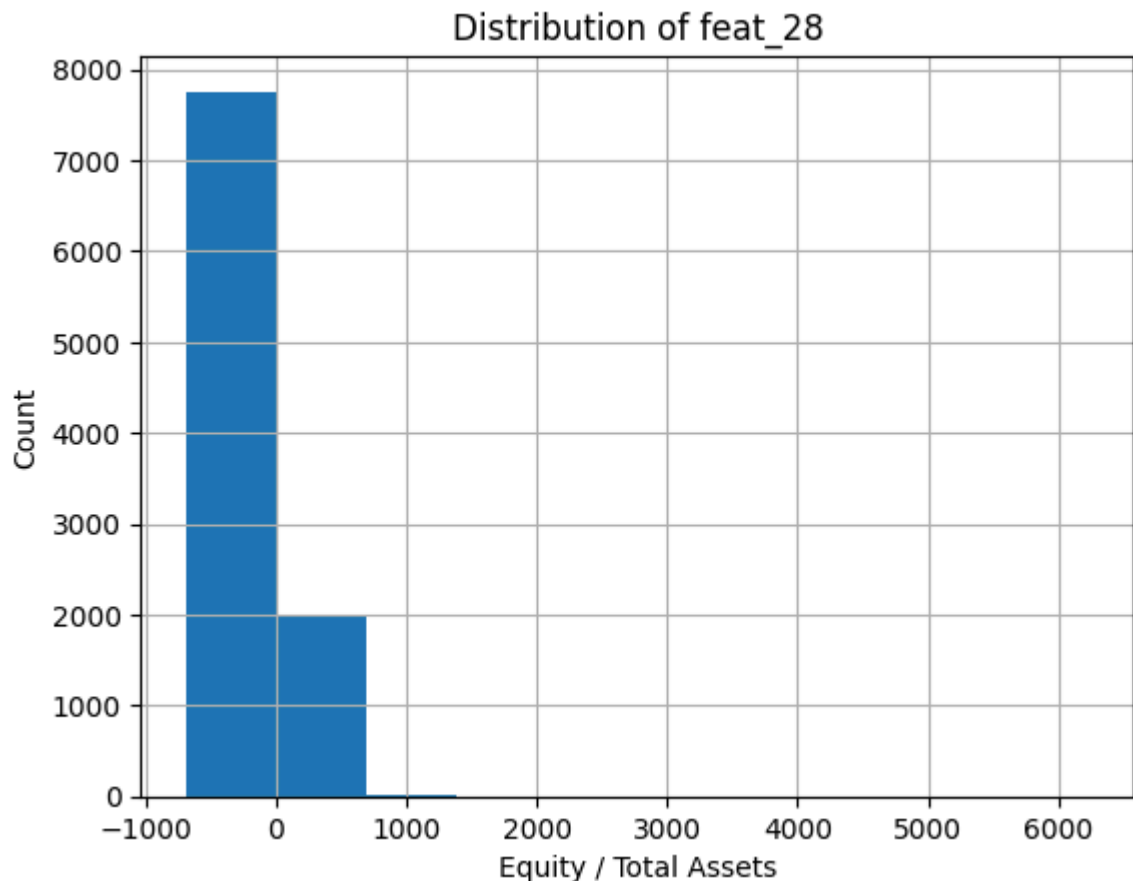
More context on "feat_27" : *Profit on operating activities* is profit that a company makes through its "normal" operations. For instance, a car company profits from the sale of its cars. However, a company may have other forms of profit, such as financial investments. So a company's *total profit* may be positive even when its profit on operating activities is negative.

Financial expenses include things like interest due on loans, and does not include "normal" expenses (like the money that a car company spends on raw materials to manufacture cars).

Task 5.2.8: Repeat the exploration you just did for "feat_27" on two other features in the dataset. Do they show the same skewed distribution? Are there large differences between bankrupt and solvent companies?

```
In [10]: # Explore another feature
df["feat_28"].hist()
plt.xlabel("Equity / Total Assets")
```

```
plt.ylabel("Count")
plt.title("Distribution of feat_28");
```



Looking at other features, we can see that they're skewed, too. This will be important to keep in mind when we decide what type of model we want to use.

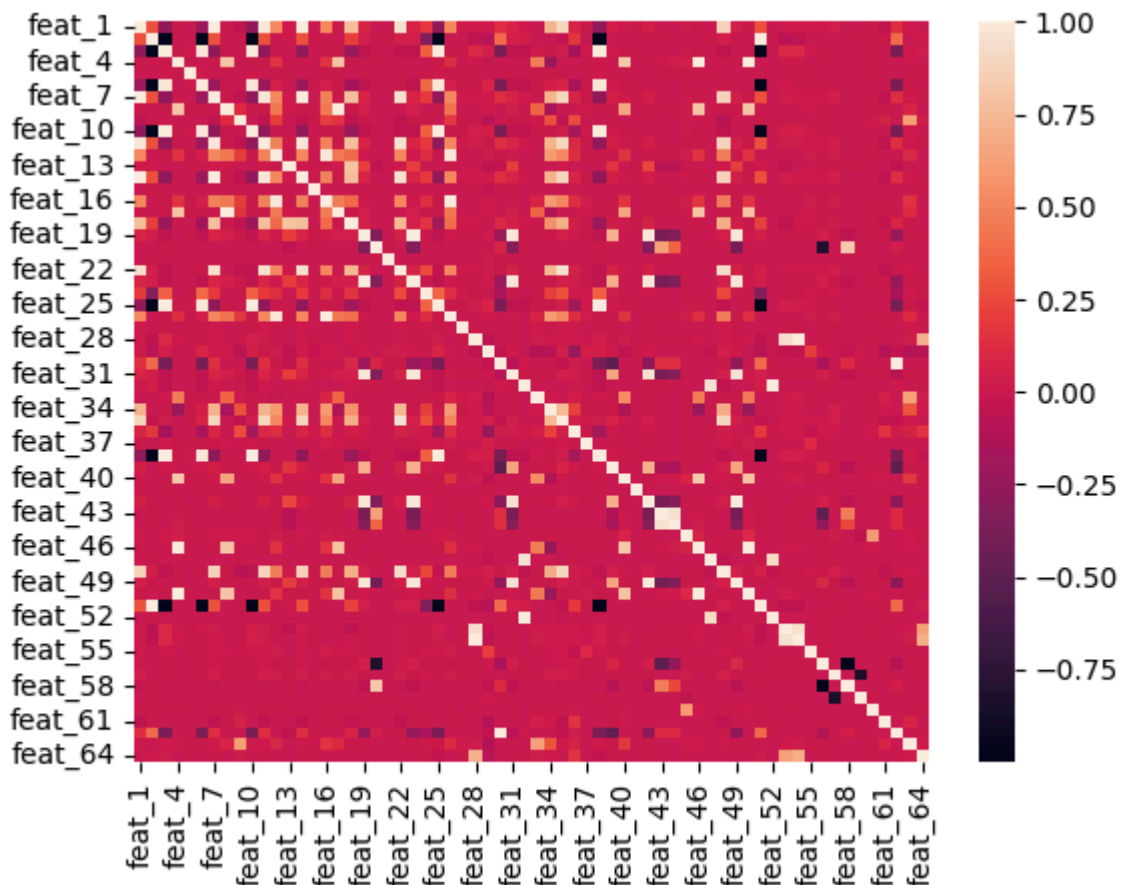
Another important consideration for model selection is whether there are any issues with multicollinearity in our model. Let's check.

```
In [ ]: VimeoVideo("694058273", h="85b3be2f63", width=600)
```

Task 5.2.9: Plot a correlation heatmap of features in `df`. Since `"bankrupt"` will be your target, you don't need to include it in your heatmap.

- [What's a heatmap?](#)
- [Create a correlation matrix in pandas.](#)
- [Create a heatmap in seaborn.](#)

```
In [11]: corr = df.drop(columns="bankrupt").corr()
sns.heatmap(corr);
```



So what did we learn from this EDA? First, our data is imbalanced. This is something we need to address in our data preparation. Second, many of our features have missing values that we'll need to impute. And since the features are highly skewed, the best imputation strategy is likely median, not mean. Finally, we have autocorrelation issues, which means that we should steer clear of linear models, and try a tree-based model instead.

Split

So let's start building that model. If you need a refresher on how and why we split data in these situations, take a look back at the Time Series module.

Task 5.2.10: Create your feature matrix `X` and target vector `y`. Your target is "bankrupt".

- [What's a feature matrix?](#)
- [What's a target vector?](#)
- [Subset a DataFrame by selecting one or more columns in pandas.](#)
- [Select a Series from a DataFrame in pandas.](#)

```
In [12]: target = "bankrupt"
X = df.drop(columns=target)
y = df[target]
print("X shape:", X.shape)
print("y shape:", y.shape)
```

```
X shape: (9977, 64)
y shape: (9977,)
```

In order to make sure that our model can generalize, we need to put aside a test set that we'll use to evaluate our model once it's trained.

Task 5.2.11: Divide your data (`X` and `y`) into training and test sets using a randomized train-test split. Your validation set should be 20% of your total data. And don't forget to set a `random_state` for reproducibility.

- [Perform a randomized train-test split using scikit-learn.](#)

```
In [13]: X_train, X_test, y_train, y_test = train_test_split(
          X, y, test_size=0.2, random_state=42
        )
print("X_train shape:", X_train.shape)
print("y_train shape:", y_train.shape)
print("X_test shape:", X_test.shape)
print("y_test shape:", y_test.shape)
```

```
X_train shape: (7981, 64)
y_train shape: (7981,)
X_test shape: (1996, 64)
y_test shape: (1996,)
```

Note that if we wanted to tune any hyperparameters for our model, we'd do another split here, further dividing the training set into training and validation sets. However, we're going to leave hyperparameters for the next lesson, so no need to do the extra split now.

Resample

Now that we've split our data into training and validation sets, we can address the class imbalance we saw during our EDA. One strategy is to resample the training data. (This will be different than the resampling we did with time series data in Project 3.) There are many to do this, so let's start with under-sampling.

```
In [ ]: VimeoVideo("694058220", h="00c3a98358", width=600)
```

Task 5.2.12: Create a new feature matrix `X_train_under` and target vector `y_train_under` by performing random under-sampling on your training data.

- [What is under-sampling?](#)
- [Perform random under-sampling using imbalanced-learn.](#)

```
In [14]: under_sampler = RandomUnderSampler(random_state=42)
X_train_under, y_train_under = under_sampler.fit_resample(X_train, y_train)
print(X_train_under.shape)
X_train_under.head()
```

```
(768, 64)
```

Out[14]:

	feat_1	feat_2	feat_3	feat_4	feat_5	feat_6	feat_7	feat_8	feat_9
company_id									
1509	0.121400	0.097238	0.74500	8.66160	25.8370	0.000000	0.158840	9.2841	
6096	0.316280	0.264920	0.65346	3.46660	55.1890	0.290140	0.316280	2.7747	
7369	0.066615	0.890270	-0.48007	0.46076	-76.3080	0.047487	0.087936	-0.0114	
5268	0.208000	0.363630	0.33875	1.93160	-5.3169	0.251870	0.258280	1.6033	
395	0.096863	0.299320	0.35393	2.42640	31.5630	0.267550	0.125130	2.3410	

5 rows × 64 columns



Note: Depending on the random state you set above, you may get a different shape for `X_train_under`. Don't worry, it's normal!

And then we'll over-sample.

In []: `VimeoVideo("694058177", h="5cef977f2d", width=600)`

Task 5.2.13: Create a new feature matrix `X_train_over` and target vector `y_train_over` by performing random over-sampling on your training data.

- [What is over-sampling?](#)
- [Perform random over-sampling using imbalanced-learn.](#)

In [15]:

```

over_sampler = RandomOverSampler(random_state=42)
X_train_over, y_train_over = over_sampler.fit_resample(X_train, y_train)
print(X_train_over.shape)
X_train_over.head()

```

(15194, 64)

Out[15]:

	feat_1	feat_2	feat_3	feat_4	feat_5	feat_6	feat_7	feat_8	feat_9
0	0.279320	0.053105	0.852030	17.0440	199.080	0.741770	0.353570	16.00600	1.2346
1	0.001871	0.735120	0.156460	1.2269	-10.837	0.000000	0.002938	0.36032	1.4809
2	0.113940	0.490250	0.077121	1.2332	-43.184	-0.000171	0.113940	1.03980	1.1649
3	0.008136	0.652610	0.148120	1.2628	29.071	0.000000	0.008136	0.53230	1.2891
4	0.045396	0.279640	0.708730	3.7656	238.120	0.000000	0.056710	2.57610	1.0169

5 rows × 64 columns



Build Model

Baseline

As always, we need to establish the baseline for our model. Since this is a classification problem, we'll use accuracy score.

```
In [ ]: VimeoVideo("694058140", h="7ae111412f", width=600)
```

Task 5.2.14: Calculate the baseline accuracy score for your model.

- [What's accuracy score?](#)
- [Aggregate data in a Series using value_counts in pandas.](#)

```
In [16]: acc_baseline = y_train.value_counts(normalize=True).max()  
print("Baseline Accuracy:", round(acc_baseline, 4))
```

Baseline Accuracy: 0.9519

Note here that, because our classes are imbalanced, the baseline accuracy is very high. We should keep this in mind because, even if our trained model gets a high validation accuracy score, that doesn't mean it's actually *good*.

Iterate

Now that we have a baseline, let's build a model to see if we can beat it.

```
In [ ]: VimeoVideo("694058110", h="dc751751bf", width=600)
```

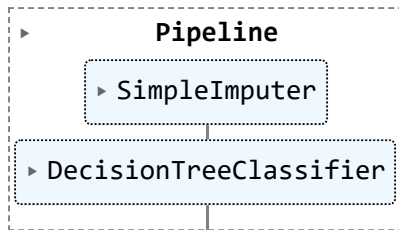
Task 5.2.15: Create three identical models: `model_reg`, `model_under` and `model_over`. All of them should use a `SimpleImputer` followed by a `DecisionTreeClassifier`. Train `model_reg` using the unaltered training data. For `model_under`, use the undersampled data. For `model_over`, use the oversampled data.

- [What's a decision tree?](#)
- [What's imputation?](#)
- [Create a pipeline in scikit-learn.](#)
- [Fit a model to training data in scikit-learn.](#)

```
In [17]: # Fit on `X_train`, `y_train`  
model_reg = make_pipeline(  
    SimpleImputer(strategy="median"),  
    DecisionTreeClassifier(random_state=42)  
)  
model_reg.fit(X_train, y_train)  
  
# Fit on `X_train_under`, `y_train_under`  
model_under = make_pipeline(  
    SimpleImputer(strategy="median"),  
    DecisionTreeClassifier(random_state=42)  
)  
model_under.fit(X_train_under, y_train_under)
```

```
# Fit on `X_train_over`, `y_train_over`
model_over = make_pipeline(
    SimpleImputer(strategy="median"),
    DecisionTreeClassifier(random_state=42)
)
model_over.fit(X_train_over, y_train_over)
```

Out[17]:



Evaluate

How did we do?

```
In [ ]: VimeoVideo("694058076", h="d57fb27d07", width=600)
```

Task 5.2.16: Calculate training and test accuracy for your three models.

- [What's an accuracy score?](#)
- [Calculate the accuracy score for a model in scikit-learn.](#)

```
In [18]: for m in [model_reg, model_under, model_over]:
    acc_train = m.score(X_train, y_train)
    acc_test = m.score(X_test, y_test)
    print("Training Accuracy:", round(acc_train, 4))
    print("Test Accuracy:", round(acc_test, 4))
```

```
Training Accuracy: 1.0
Test Accuracy: 0.9359
Training Accuracy: 0.7421
Test Accuracy: 0.7104
Training Accuracy: 1.0
Test Accuracy: 0.9344
```

```
In [19]: acc_train
```

Out[19]: 1.0

```
In [20]: acc_test
```

Out[20]: 0.9343687374749499

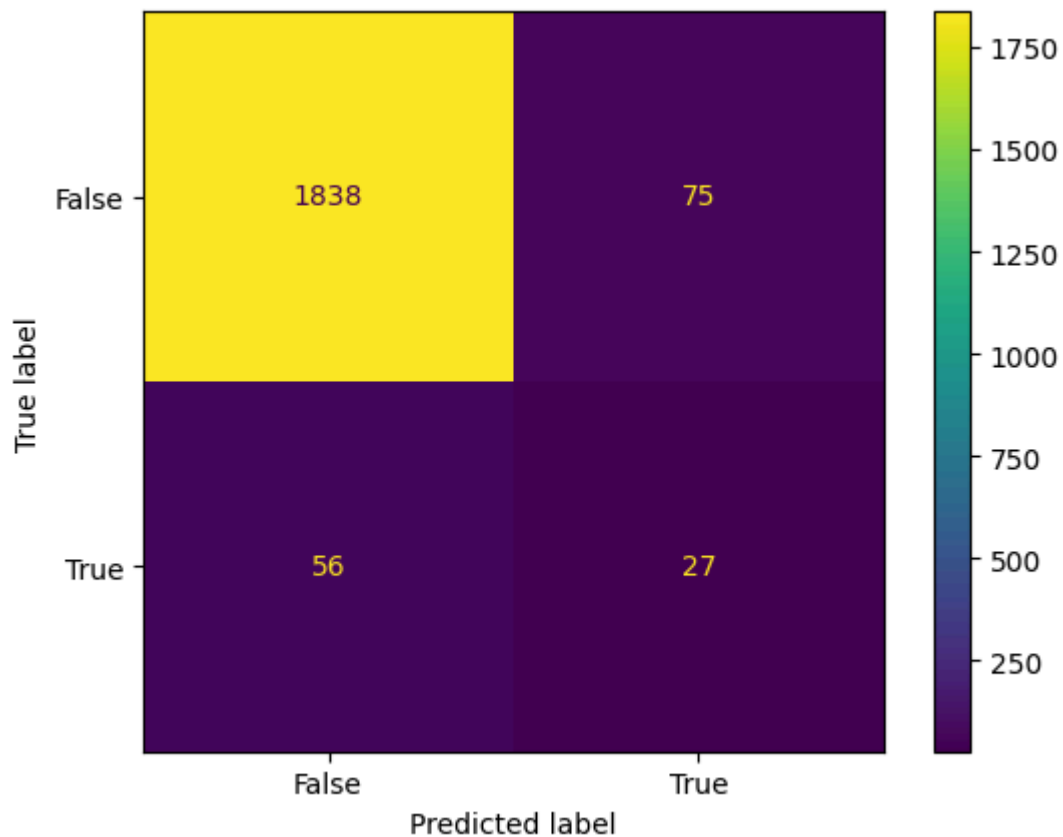
As we mentioned earlier, "good" accuracy scores don't tell us much about the model's performance when dealing with imbalanced data. So instead of looking at what the model got right or wrong, let's see how its predictions differ for the two classes in the dataset.

```
In [ ]: VimeoVideo("694058022", h="ce29f57dee", width=600)
```

Task 5.2.17: Plot a confusion matrix that shows how your best model performs on your validation set.

- [What's a confusion matrix?](#)
- [Create a confusion matrix using scikit-learn.](#)

```
In [21]: # Plot confusion matrix
ConfusionMatrixDisplay.from_estimator(model_over, X_test, y_test);
```



In this lesson, we didn't do any hyperparameter tuning, but it will be helpful in the next lesson to know what the depth of the tree `model_over` .

```
In [ ]: VimeoVideo("694057996", h="73882663cf", width=600)
```

Task 5.2.18: Determine the depth of the decision tree in `model_over` .

- [What's a decision tree?](#)
- [Access an object in a pipeline in scikit-learn.](#)

```
In [22]: depth = model_over.named_steps["decisiontreeclassifier"].get_depth()
print(depth)
```

33

Communicate

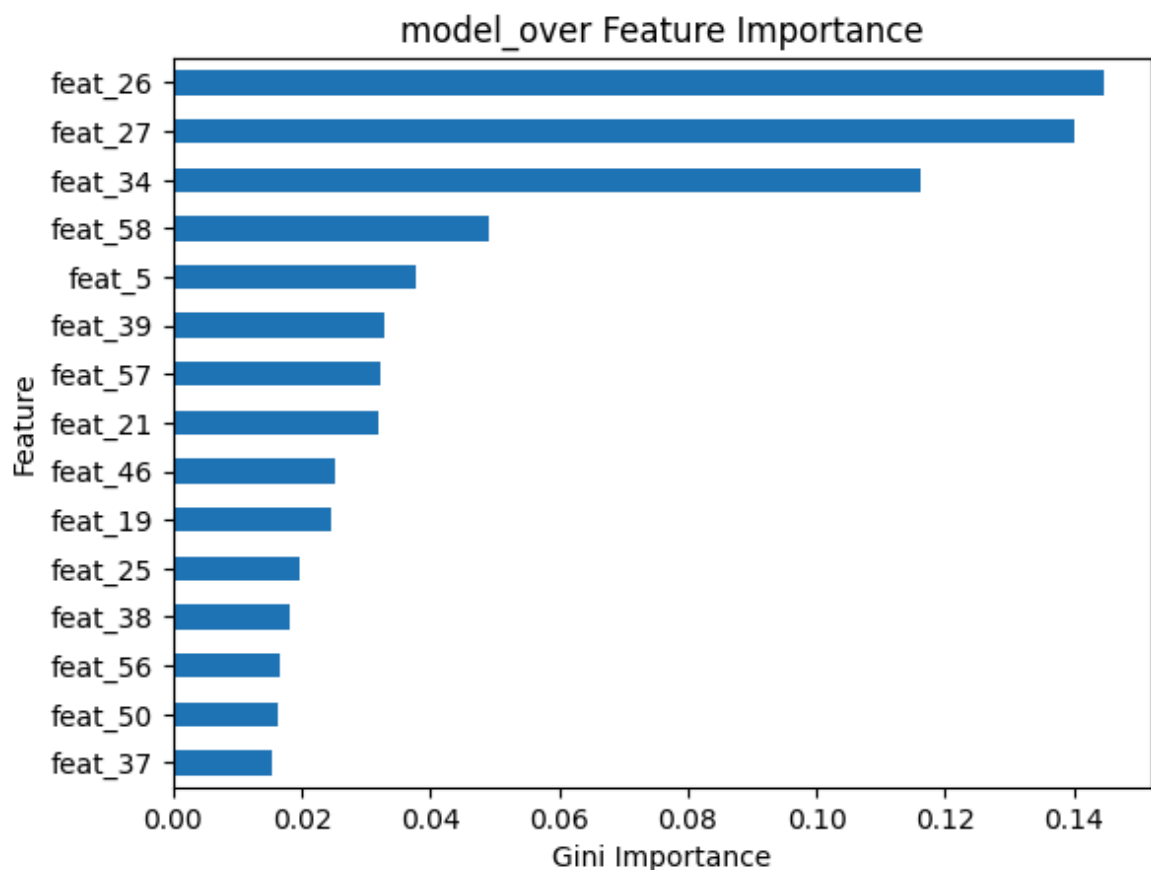
Now that we have a reasonable model, let's graph the importance of each feature.

```
In [ ]: VimeoVideo("694057962", h="f60aa3b614", width=600)
```

Task 5.2.19: Create a horizontal bar chart with the 15 most important features for `model_over`. Be sure to label your x-axis "Gini Importance".

- [What's a bar chart?](#)
- [Access an object in a pipeline in scikit-learn.](#)
- [Create a bar chart using pandas.](#)
- [Create a Series in pandas.](#)

```
In [23]: # Get importances
importances = model_over.named_steps["decisiontreeclassifier"].feature_importances_
# Put importances into a Series
feat_imp = pd.Series(importances, index=X_train_over.columns).sort_values()
# Plot series
feat_imp.tail(15).plot(kind="barh")
plt.xlabel("Gini Importance")
plt.ylabel("Feature")
plt.title("model_over Feature Importance");
```



There's our old friend "feat_27" near the top, along with features 34 and 26. It's time to share our findings.

Sometimes communication means sharing a visualization. Other times, it means sharing the actual model you've made so that colleagues can use it on new data or deploy your model into production. First step towards production: saving your model.

```
In [ ]: VimeoVideo("694057923", h="85a50bb588", width=600)
```

Task 5.2.20: Using a context manager, save your best-performing model to a file named `"model-5-2.pkl"`.

- What's serialization?
- Store a Python object as a serialized file using pickle.

```
In [24]: # Save your model as `model-5-2.pkl`
with open("model-5-2.pkl", "wb") as f:
    pickle.dump(model_over, f)
```

```
In [ ]: VimeoVideo("694057859", h="fec8f9e54", width=600)
```

Task 5.2.21: Make sure you've saved your model correctly by loading `"model-5-2.pkl"` and assigning to the variable `loaded_model`. Once you're satisfied with the result, run the last cell to submit your model to the grader.

- Load a Python object from a serialized file using pickle.

```
In [25]: # Load `model-5-2.pkl`
with open("model-5-2.pkl", "rb") as f:
    loaded_model = pickle.load(f)
print(loaded_model)
```

```
Pipeline(steps=[('simpleimputer', SimpleImputer(strategy='median')),
                 ('decisiontreeclassifier',
                  DecisionTreeClassifier(random_state=42))])
```